

(1001CJA621620035)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)

UNIT TEST

27-09-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C,D	A,C	D	A,B,C	A,B	D	A,C	A,C,D	A,B,D	A,B,D
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	1	7	3	2	2	5	5	9		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		P,Q,R,S,T	P,Q,R,S,T	P,S,T	P,Q,R,S,T		P	P	P	R,T	

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	B,C	A,B,D	A,C	A,B,C	A,B	A,C,D	B,C	A,B,C,D	A,B,C
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	1	5	2	1	5	9	7	5		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		P,Q,R	S	P,S	P,T		Q	T	P	T	

PART-3 : MATHEMATICS

PART - C: MATHEMATICS											
SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B,D	A,B	A,B	A,C,D	A,B	A,C	A,B	A,C,D	A,D	A,B
SECTION-III	Q.	1	2	3	4	5	6	7	8		
	A.	1	2	2	3	4	1	8	5		
SECTION-IV	Q.1	A	B	C	D	Q.2	A	B	C	D	
		R	S	Q	P		S	S	P	R	

(1001CJA621620036)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Main)

UNIT TEST

27-09-2020

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	A	A	A	A	A	C	D	C	A
SECTION-III	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	C	B	C	D	A	C	A	D	D
SECTION-II	Q.	1	2	3	4	5					
	A.	3.00	0.75	2.00	6.00	9.00					

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	A	B	B	D	C	D	D	A
SECTION-III	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	B	C	C	B	A	D	D	C	B
SECTION-II	Q.	1	2	3	4	5					
	A.	27.00	5.00	6.00	3.00	3.00					

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	A	A	C	C	C	B	C	B	B
SECTION-III	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	A	B	B	A	C	B	C	C	D
SECTION-II	Q.	1	2	3	4	5					
	A.	1.00	64.00	0.25	3.00	3.50					

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**PAPER-1****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (A, B, C, D)**

Sol. For node $10\pi x = \frac{(2n+1)\pi}{2}$

$$\Rightarrow x = \frac{(2n+1)}{20} \text{ \& } n = 1$$

$$\text{for anti node } 10\pi x = n\pi \Rightarrow x = \frac{n}{10} \text{ \& } n = 3$$

$$k = 10\pi = \frac{2\pi}{\lambda} \Rightarrow \lambda = 0.2\text{m}, \omega = 50\pi \Rightarrow f = 25$$

$$\text{speed} = f\lambda = 5 \text{ m/s (A,B,C,D)}$$

2. **Ans. (A,C)**

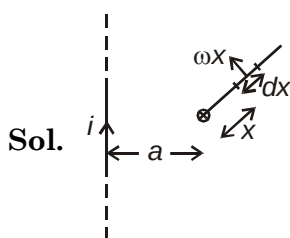
Sol. $\mu = \frac{m}{2\ell}$

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{Mg}{m/2\ell}}$$

$$\text{also } k\ell = Mg$$

$$\text{Mass of block} = \frac{k\ell}{g} \text{ (A)}$$

$$t = \frac{2\ell}{v} = 2\ell \sqrt{\frac{m}{Mg2\ell}} = \sqrt{\frac{2m}{k}} \text{ (A,C)}$$

3. **Ans. (D)**

Induced emf in the small element dx is

$$\frac{\mu_0 i dx \omega x}{2\pi(a+x \cos \theta)}$$

$$\text{So, total induced emf} = \int_0^{\ell} \frac{\mu_0 i \omega x}{2\pi(a+x \cos \theta)} dx$$

which comes out to be

$$\frac{\mu_0 i \omega}{2\pi \cos^2 \theta} \left[\ell \cos \theta - a \ln \left| \frac{a+\ell \cos \theta}{a} \right| \right]$$

4. **Ans. (A,B,C)**

Sol. $2 \frac{di}{dt} = 10 - 2i$

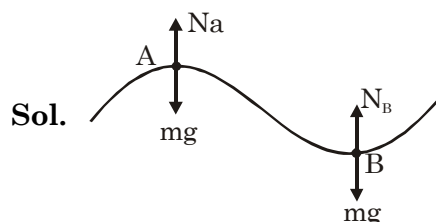
$$\frac{di}{dt} + i = 5$$

$$\therefore i = 5(1 - e^{-t})$$

$$\text{at } t = \ln 2, i = 5 \left(1 - \frac{1}{2} \right) = 2.5 \text{ A}$$

$$\therefore U = \frac{1}{2} Li^2 = 6.25 \text{ J}$$

$$\frac{dU}{dt} = \frac{1}{2} \cdot L \cdot 2i \frac{di}{dt} = Li \frac{di}{dt} = 2 \times 2.5 \times 2.5 = 12.5 \text{ J/s}$$

5. **Ans. (A,B)**

$$mg - N_A = \frac{mv^2}{R} \quad N_B - mg = \frac{mv^2}{2R}$$

$$N_A = mg + \frac{mv^2}{R} \quad N_D = mg + \frac{mv^2}{2R}$$

6. Ans. (D)

Sol. $r_{\text{rms}} = \sqrt{\frac{\int i^2 dt}{\int dt}} \quad i = \frac{2i_0}{T} t$

$$= \left[\frac{4i_0^2 \left[\frac{t^3}{3} \right]_0^{T/2}}{\frac{T}{2}} \right]^{1/2} = \frac{i_0}{\sqrt{3}} \quad (\text{D})$$

7. Ans. (A, C)

Sol. $(V_L - V_C) = 50$

$$V_R^2 + (V_L - V_C)^2 = 130^2 \Rightarrow V_R^2 = 130^2 - 50^2$$

$$\therefore \text{power factor} = \frac{V_R}{V} = \frac{120}{130} = \frac{12}{13}$$

$$V_R^2 + V_L^2 = 150^2$$

$$\Rightarrow V_L = \sqrt{150^2 - 120^2} = 90 \text{ V}$$

$$V_L - V_C = 50 \Rightarrow V_C = 90 - 50 = 40 \text{ V}$$

8. Ans. (A, C, D)

Sol. Velocity of first ball just before collision

$$= \sqrt{2 \times \frac{F}{m} \times d}$$

$$(\because v^2 = u^2 + 2as)$$

$\therefore$ Collision is elastic masses are same, hence after collision, velocity will be exchanged.

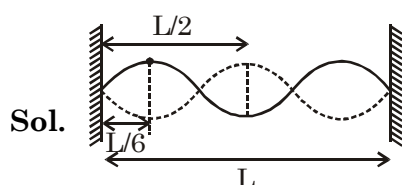
$$v_{\text{av}} = \frac{u + v}{2} = \frac{1}{2} \sqrt{\frac{2Fd}{m}}$$

9. Ans. (A, B, D)

Sol. For same speed $\Delta x = \frac{n\lambda}{2}$

& for same velocity $\Delta x = n\lambda$

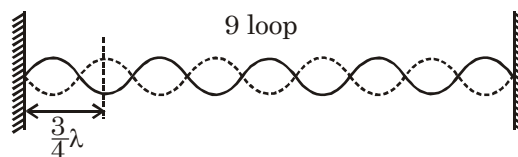
10. Ans. (A, B, D)



From given information

3 loop or it is 2nd overtone

at $L/2$ there is antinode



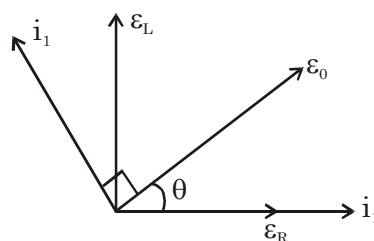
$$9 \frac{\lambda}{2} = \ell$$

$$\lambda = \frac{2\ell}{9}$$

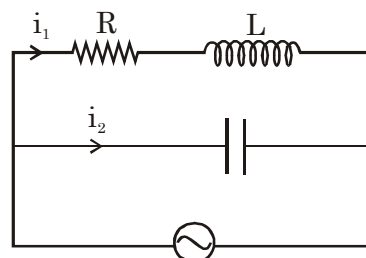
$$\text{distance} = \frac{3}{4} \left(\frac{2\ell}{9} \right) = \frac{\ell}{6}$$

SECTION-III

1. Ans. 1



Sol.



for net current to be in same phase

$$i_1 \sin \theta = i_2$$

$$\frac{\epsilon_0 x_L}{R^2 + x_L^2} = \frac{\epsilon_0}{x_C}$$

$$x_L x_C = R^2 + x_L^2$$

$$\frac{L}{C} = R^2 + \omega^2 L^2$$

$$\Rightarrow \omega \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

$$f = \frac{1}{2\pi} \sqrt{\frac{1}{LC} - \frac{R^2}{L^2}}$$

2. Ans. 7

Sol. $a = \frac{qE}{m}$

$$E = \frac{R^2}{2r} \frac{dB}{dt} = \frac{1}{2} (8t - 2) = 7$$

$$x = 7$$

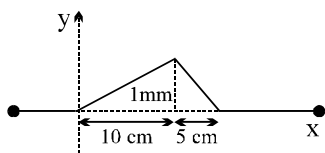
3. Ans. 3

Sol. at given time $y = m_1 x$ $0 < x < 0.1$ m
 $= m_2 x + C$ $0.1 \text{ m} < x < 0.15$ m

$$m_1 = \frac{10^{-3}}{0.1} = 10^{-2} \quad \left(V_\omega = \sqrt{\frac{T}{\mu}} \right)$$

$$m_2 = \frac{10^{-3}}{0.05} = 2 \times 10^{-2}$$

$$dK = \frac{1}{2} dm V_p^2 = \frac{1}{2} \mu dx \left(-V_\omega \frac{\partial y}{\partial x} \right)^2$$



$$\int dK = \frac{1}{2} \mu V_\omega^2 \left[m_1^2 \int_0^{0.1} dx + m_2^2 \int_{0.1}^{0.15} dx \right]$$

$$K = \frac{1}{2} T [m_1^2 (0.1) + m_2^2 (0.5)]$$

$$K = \frac{1}{2} T [10^{-4} (0.1) + 4 \times 10^{-4} (0.5)]$$

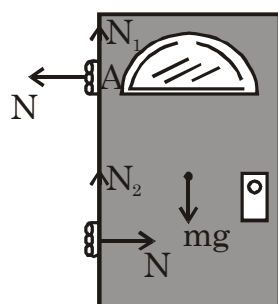
$$= \frac{10}{2} [10^{-4}] (0.3) = 0.15 \text{ mJ}$$

4. Ans. 2

Sol. $IL^2 (\hat{i} + \hat{j}) = \sqrt{2} IL^2$

$$x = \sqrt{2}$$

5. Ans. 2



Sol.

Balancing torque w.r.t point A

$$N \cdot 150 = mg \times \frac{75}{2}$$

$$N = \frac{mg}{4} = \frac{8 \times 10}{4} = 20$$

6. Ans. 5

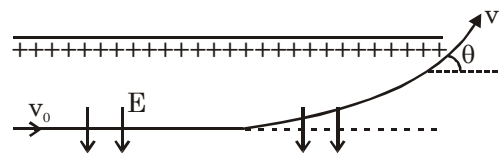
Sol. $H = i^2 R = \left(\frac{V_0}{\sqrt{(R+r)^2 + (\omega L)^2}} \right)^2 \times R$ for

maximum value of H solve $\frac{dH}{dR} = 0$

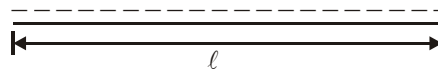
On solving $R = 5\Omega$

7. Ans. 5

8. Ans. 9



Sol.



Since, electron is accelerated through a potential difference V, therefore, its initial velocity v_0 is given by

$$\frac{1}{2} m v_0^2 = eV$$

$$\text{Or } v_0 = \sqrt{\frac{2eV}{m}} \quad \dots(i)$$

Since, initial velocity is parallel to plates or normal to the direction of electric field, therefore, component of velocity parallel to plates remains constant as v_0 .

Hence, time taken by the electron to cross

$$\text{electric field is } t_0 = \frac{1}{v_0}$$

Now consider motion of electron, normal to plates.

At some instant t, its acceleration

$$= \frac{eE}{m} = \frac{eat}{m}$$

Let velocity component normal to plates be v_y

$$\therefore \frac{d}{dt} v_y = \frac{eat}{m}$$

$$\therefore \int_0^{v_y} dv_y = \frac{eat}{m} \int_0^{t_0} t dt$$

$$\text{or } v_y = \frac{ea}{2m} t_0^2 = \frac{a\ell^2}{4V} \quad \dots(ii)$$

If q is angular deviation of electron from its initial direction of motion.

The pitch of its helical path.

$$p = \frac{2\pi m}{eB} v \cos(90 - \theta)$$

$$\therefore p = \frac{2\pi m}{eB} v \sin \theta = \frac{2\pi m}{eB} v_y$$

$$\text{or } p = \frac{\pi m a \ell^2}{2eBV}$$

$$p = 9\text{mm}$$

SECTION-IV

1. Ans. (A)-(P,Q,R,S,T); (B)-(P,Q,R,S,T); (C)-(P,S,T); (D)-(P,Q,R,S,T)
2. Ans. (A)-(P); (B)-(P); (C)-(P); (D)-(R,T)

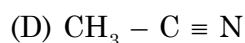
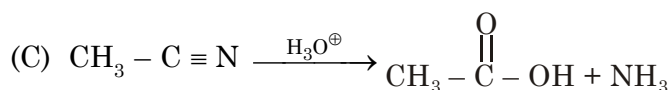
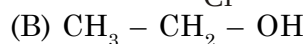
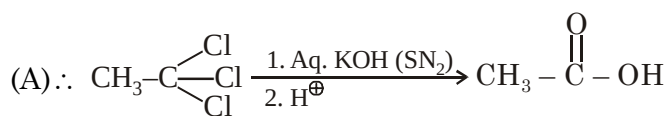
PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(A, C, D)
2. Ans.(B, C)
3. Ans. (A, B, D)
4. Ans. (A,C)

Sol.

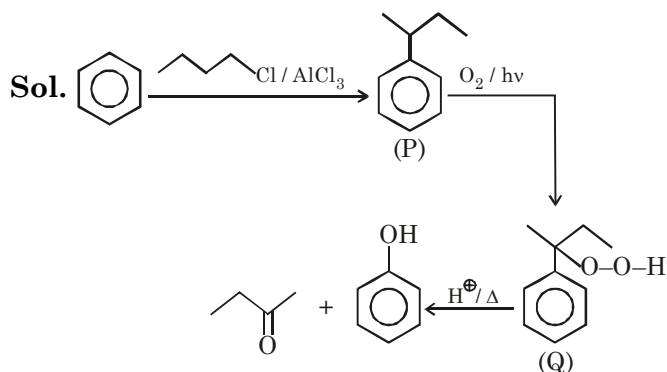


5. Ans. (A,B,C)

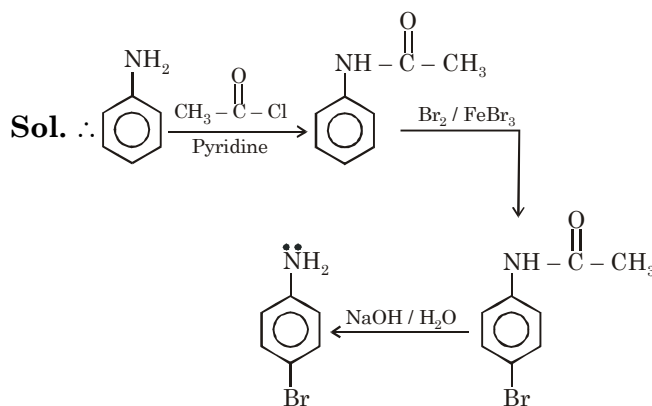
Sol. $\therefore$ (A) Etard reaction

- (B) Mild oxidising agent
- (C) Rosenmund reduction
- (D) Haloform reaction

6. Ans. (A, B)



7. Ans. (A,C,D)



8. Ans.(B,C)

$$(\text{pH})_{\text{eq. percent}} = \frac{1}{2}(\text{pK}_a - \log C) = \frac{1}{2}(5 + 1) = 3$$

9. Ans.(A,B,C,D)

10. Ans (A,B,C)

$$\begin{aligned} \text{(A) Eq. of } \text{MnO}_4^- &= \text{mole} \times n_f \\ &= 1 \times 5 \end{aligned}$$

$$\text{Eq. of } \text{Fe}^{2+} = 10 \times 1 = 10$$

$$\text{Eq. of } \text{MnO}_4^- < \text{Eq. of } \text{Fe}^{2+}$$

$$\text{(D) Eq. of } \text{Cu}_2\text{S} = 2 \times 8 = 16$$

$$\text{Eq. of } \text{Cr}_2\text{O}_7^{2-} = 2.66 \times 6 = 16$$

$$\text{Eq. of } \text{Cu}_2\text{S} = \text{Eq. of } \text{Cr}_2\text{O}_7^{2-}$$

SECTION-III

1. Ans.(1)

$$\frac{K_{sp}}{K_a} = \frac{s^2}{[\text{H}^+]}$$

$$\frac{10^{-12}}{10^{-5}} = \frac{s^2}{10^{-3}}$$

$$s_1 = 10^{-5}M$$

$$s_2 = \sqrt{K_{sp}} = 10^{-6}M$$

$$\frac{s_1}{s_2} = 10$$

2. Ans.(5)

3. Ans.(2)

4. Ans.(1)

5. Ans.(5)

6. Ans.(9)

7. Ans. (7)

Sol. Carbyl amine reaction gives foul smell with 1° amine and aniline.

8. Ans. (5)

Sol. (+M, +H, +I) groups.

SECTION-IV

1. Ans.(A)→(P,Q,R); (B)→(S); (C)→(P,S); (D)→(P,T)

2. Ans.(A)→(Q); (B)→(T); (C)→(P); (D)→(T)

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (B,D)

$$f(x) = x \cot^{-1} x$$

$$f'(x) = \cot^{-1} x - \frac{x}{1+x^2}$$

$$f''(x) = -\frac{2}{(1+x^2)^2}$$

$f'(x)$ is strictly decreasing $\forall x \in (0, \infty)$

$$x < \infty$$

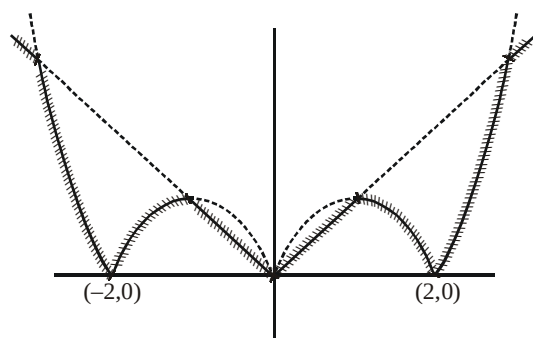
$$f'(x) > \lim_{x \rightarrow \infty} f'(x) \Rightarrow f'(x) > 0$$

$$\cot^{-1} x > \frac{x}{1+x^2}$$

2. Ans. (A,B)

$$f(x) = \max \{(|x|^2 - 2|x|), |x|\}$$

$$g(x) = \min \{(|x|^2 - 2|x|), |x|\}$$



$f(x)$ non differentiable at 5 points

$g(x)$ non differentiable at 7 points

3. Ans. (A,B)

$$f(x) = \int_0^x f(t) \cos t dt - \int_0^x \cos(-t) dt$$

$$f'(x) = f(x) \cos x - \cos x$$

$$f'(x) - f(x) \cos x = -\cos x, \text{ I.F.} = e^{-\sin x}$$

$$e^{-\sin x} \cdot f(x) = \int e^{-\sin x} \cdot (-\cos x) dx$$

$$e^{-\sin x} \cdot f(x) = e^{-\sin x} + c$$

$$f(x) = 1 + ce^{\sin x}, \text{ Now } f(0) = 0 \Rightarrow c = -1$$

$$\Rightarrow f(x) = 1 - e^{\sin x}$$

4. Ans. (A,C,D)

$$\text{Let } f(x) = ax^3 + bx^2 + cx + d$$

$f(x)$ is min at $x = -2$ and max at $x = 2$

$$\Rightarrow a < 0 \Rightarrow a^2 - a - 6 = 0$$

$$a = 3, -2 \text{ (} a = 3 \text{ is rejected)}$$

$$f(x) = -2x^3 + bx^2 + cx + d$$

$$f'(x) = -6x^2 + 2bx + c = -6(x^2 - 4)$$

$$b = 0, c = 24, d \in \mathbb{R}$$

5. Ans. (A,B)

$$\vec{a} \cdot \vec{b} = 0, \quad \vec{c} \cdot \vec{a} = 1, \quad \vec{c} \cdot \vec{b} = \frac{1}{2}$$

$$\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix} = \begin{bmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{bmatrix}$$

$$\begin{bmatrix} \vec{a} & \vec{b} & \vec{c} \end{bmatrix}^2 = \begin{vmatrix} 4 & 0 & 1 \\ 0 & 1 & \frac{1}{2} \\ 1 & \frac{1}{2} & 1 \end{vmatrix} = 2$$

$$\vec{c} = x\vec{a} + y\vec{b} + z(\vec{a} \times \vec{b})$$

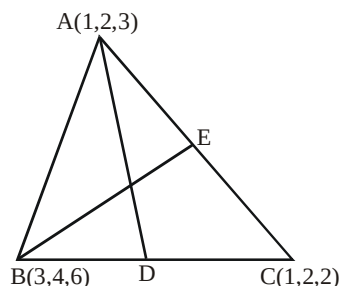
Dot with $\vec{a}, \vec{b}$ and $\vec{a} \times \vec{b}$ we get

$$\vec{c} \cdot \vec{a} = 4x = 1, \quad \vec{c} \cdot \vec{b} = y = \frac{1}{2}$$

$$[\vec{a} \quad \vec{b} \quad \vec{c}] = z|\vec{a} \times \vec{b}|^2 \Rightarrow 4z = \pm\sqrt{2}$$

$$z = \pm \frac{1}{2\sqrt{2}}, y = \frac{1}{2}, x = \frac{1}{4}$$

6. Ans. (A,C)



$$D \equiv \left(\frac{\alpha+1}{2}, 3, \frac{\beta+6}{2} \right)$$

$$\overrightarrow{AD} = \left(\frac{\alpha-1}{2} \hat{i} + \hat{j} + \frac{\beta}{2} \hat{k} \right)$$

$$\frac{\alpha-1}{2} = 1 = \frac{\beta}{2} \Rightarrow (\alpha, \beta) \equiv (3, 2)$$

$$D \equiv (2, 3, 4)$$

median through B :

$$B \equiv (3, 4, 6) \text{ and } E \left(1, 2, \frac{5}{2} \right)$$

$$\frac{x-3}{4} = \frac{y-4}{4} = \frac{z-6}{7}$$

7. Ans. (A,B)

$$A^n = A^{n-2} + A^2 - I$$

$$\text{Put, } n=4 \quad A^4 - A^2 = A^2 - I$$

$$A^4 = 2A^2 - I$$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 0 \\ 2 & 0 & 2 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}$$

8. Ans. (A,C,D)

$$\lim_{x \rightarrow -1} \frac{\sin(x^3 + bx^2 + cx + d)}{(\log_e^2(x+2))(\sqrt{2+x}-1)}$$

$x = -1 + t$ and rationalize we get

$$2 \lim_{t \rightarrow 0} \frac{\sin(t^3 + t^2(b-3) + t(3-2b+c) + (-1+b-c+d))}{t^3}$$

$$b+d-1-c=0, 3-2b+c=0, b-3=0, \ell=2$$

$$b=3, c=3, d=1, \ell=2$$

9. Ans. (A,D)

$$a, b, c \in W \text{ and } b^2 = ac$$

$$a+b+c=13$$

Two possibility are $(a, b, c) \equiv (1, 3, 9)$ and $(9, 3, 1)$

$$T_{r+1} = \frac{13! 2^a \cdot (-1)^b \cdot (3)^c}{a! b! c!} \cdot x^a y^b z^c$$

$$\text{Ans. } \frac{13! 2^1 \cdot (-1)^3 \cdot 3^9}{9! 3! 1!} \text{ or } \frac{13! 2^9 \cdot (-1)^3 \cdot 3^1}{9! 3! 1!}$$

10. Ans. (A,B)

$$\text{Put } x=0 \Rightarrow a_0 = 1$$

$$\text{Put } x=1, 10^{15} = a_0 + a_1 + a_2 + \dots + a_{45} \dots (1)$$

$$\text{Put } x=-1,$$

$$-2^{15} = a_0 - a_1 + a_2 - a_3 + \dots - a_{45} \dots (2)$$

$$(1) + (2) \text{ gives } \quad (1) - (2) \text{ gives}$$

$$\sum_{r=0}^{22} a_{2r} = \frac{10^{15} - 2^{15}}{2}, \quad \sum_{r=0}^{22} a_{2r+1} = \frac{10^{15} + 2^{15}}{2}$$

differentiate and put $x=0$ we get $a_1 = 30$

SECTION-III

1. Ans. 1

$$A = \begin{bmatrix} 2 & 4 \\ -1 & -2 \end{bmatrix} \Rightarrow A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$|I + 2A + 3A^2 + \dots + 101A^{100}| = |I + 2A|$$

$$(I + 2A) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 8 \\ -2 & -4 \end{bmatrix} = \begin{bmatrix} 5 & 8 \\ -2 & -3 \end{bmatrix}$$

$$|I + 2A| = 1$$

2. Ans. 2

$$\int \cos^5 x \cdot \cos x dx = \int (1 - \sin^2 x)^2 \cos x dx$$

$$\text{put } \sin x = t$$

$$\int (1 - t^2)^2 dt = \frac{t^5}{5} - \frac{2t^3}{3} + t + c$$

$$= \frac{\sin^5 x}{5} - \frac{2\sin^3 x}{3} + \sin x + c$$

3. Ans. 2

$$I = \int_0^{2021} (f'(x) + f'(-x)) \left(\frac{1}{(2021)^x + 1} + \frac{1}{(2021)^{-x} + 1} \right) dx$$

$$\int_0^{2021} (f'(x) + f'(-x)) dx = (f(x) - f(-x)) \Big|_0^{2021}$$

$$I = f(2021) - f(-2021)$$

$$f(x) = x^{100} - 2x^{98} + x^{54} + 3x^{28} + \frac{x}{2021} - 1000$$

$$f(x) - f(-x) = \frac{2x}{2021}$$

$$f(2021) - f(-2021) = 2 = I$$

4. Ans. 3

$$y^2 = x \quad \left(a = \frac{1}{4} \right)$$

$$x^2 = 9y \quad \left(b = \frac{9}{4} \right)$$

$$A = \frac{16ab}{3} = 3$$

5. Ans. 4

$$\frac{x-1}{2} = \frac{y}{-1} = \frac{z}{2} \text{ lies in both the planes}$$

$$(x, y, z) \equiv (1 + 2t, -t, 2t)$$

$$x - y + z = 2 \Rightarrow (1 + 2t) + t + 2t - 2 \Rightarrow t = \frac{1}{5}$$

$$\lambda x + 3z + 5 = 0 \Rightarrow \lambda(1 + 2t) + 3(2t) + 5 = 0$$

$$\frac{7\lambda}{5} + \frac{6}{5} + 5 = 0 \Rightarrow \lambda = -\frac{31}{7}$$

6. Ans. 1

$$2x - 5y + 4z + 7 + \lambda(x + 2y + 3z + 5)$$

$$x(\lambda + 2) + y(2\lambda - 5) + z(3\lambda + 4) + (5\lambda + 7) = 0$$

this plane is parallel to line

$$\Rightarrow (\lambda + 2) + (2\lambda - 5) + 2(3\lambda + 4) = 0$$

$$9\lambda + 5 = 0 \Rightarrow \lambda = -\frac{5}{9}$$

$$\text{plane : } 13x - 55y + 21z + 38 = 0$$

$$z\text{-intercept} = -\frac{38}{21} = \lambda$$

7. Ans. 8

$yt = x + 4t^2$ is tangent to

$$x^2 = 128y$$

$$x^2 = 128 \left(\frac{x}{t} + 4t \right) \Rightarrow x^2 - \frac{128x}{t} - 128 \times 4t = 0$$

$$D = 0$$

$$\left(\frac{128}{t} \right)^2 + 16 \times 128t = 0$$

$$t^3 + 8 = 0 \Rightarrow t = -2$$

$$-2y = x + 16$$

$$\lambda = -8 \Rightarrow |\lambda| = 8$$

8. Ans. 5

$$P \equiv (4 - 5\alpha, \alpha)$$

$$\frac{x(4 - 5\alpha)}{5} + \frac{y\alpha}{3} = 1$$

$$3x(4 - 5\alpha) + 5\alpha y = 15$$

$$(12x - 15) + \alpha(-15x + 5y) = 0$$

$$x = \frac{5}{4} \text{ and } y = 3x$$

$$(x, y) \equiv \left(\frac{5}{4}, \frac{15}{4} \right) = (\alpha, \beta)$$

SECTION-IV

1. Ans. (A)→(R); (B)→(S); (C)→(Q); (D)→(P)

$$P(t_1) \text{ and } Q(t_2)$$

$$(t_1 t_2, t_1 + t_2) \equiv (-15, -2)$$

$$t_1 t_2 = -15 \text{ and } t_1 + t_2 = -2$$

$$\Rightarrow (t_1, t_2) \equiv (-5, 3) \text{ and } (3, -5)$$

$$\text{given } t_1 > t_2 \Rightarrow (t_1, t_2) \equiv (3, -5)$$

$$P \equiv (9, 6), Q(25, -10), R(-15, -2)$$

Point of intersection of normal at P and Q

$$T = (t_1^2 + t_2^2 + t_1 t_2 + 2, -t_1 t_2(t_1 + t_2))$$

$$T \equiv (21, -30)$$

$$S = \frac{S_1^{3/2}}{2} = 256$$

Co-ordinate of circumcentre = midpoint of RT
 $\equiv (3, -16) = (x_1, y_1)$

$$|x_1 y_1| = 48$$

$$PQ = 16\sqrt{2}$$

2. Ans. (A)→(S); (B)→(S); (C)→(P);
 (D)→(R)

$$(A) \quad PS_1 + PS_2 = 2a = 8$$

$$(B) \quad T = S_1 \Rightarrow \frac{xx_1}{16} + \frac{yy_1}{25} = \frac{x_1^2}{16} + \frac{y_1^2}{25}$$

$$(x_1 y_1) = (0, 3) \Rightarrow \frac{3y}{25} = \frac{9}{25} \Rightarrow y = 3$$

$$\text{solve } y = 3 \text{ and } \frac{x^2}{16} + \frac{y^2}{25} = 1$$

$$\Rightarrow x = \pm \frac{16}{5} \quad P \equiv \left(\frac{16}{5}, 3\right), Q = \left(-\frac{16}{5}, 3\right)$$

$$PQ = \frac{32}{5} = \frac{4k}{5} \Rightarrow k = 8$$

$$(C) \quad \frac{x^2}{4} + y^2 = 1$$

$$y = mx \pm \sqrt{4m^2 + 1}, \quad p = r$$

$$\left| \frac{\sqrt{4m^2 + 1}}{\sqrt{m^2 + 1}} \right| = \sqrt{3} \Rightarrow 4m^2 + 1 = 3m^2 + 3$$

$$\Rightarrow m^2 = 2$$

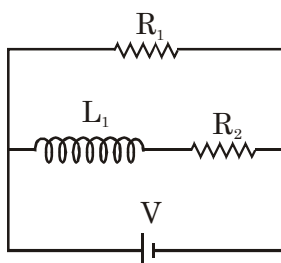
$$(D) \quad 9x^2 + 5(y - 3)^2 = 45$$

$$\frac{x^2}{5} + \frac{(y - 3)^2}{9} = 1$$

$$L(LR) = \frac{2 \times 5}{3} = \ell, e = \sqrt{1 - \frac{5}{9}} = \frac{2}{3}$$

$$e + \ell = 4$$

$$3(e + \ell) = 12$$

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & I]**PAPER-2****PART-1 : PHYSICS****SOLUTION****SECTION-I**1. **Ans. (C)****Sol.** Current will induced in such a way to oppose the change.2. **Ans. (A)****Sol.** After the battery is disconnected, current will decay as

$$I = I_0 e^{-(R_1 + R_2) \frac{t}{L}}$$

$$I_0 = \frac{V}{R_2}$$

So, total heat generated,

$$\Delta Q = \int_0^{\infty} I^2 (R_1 + R_2) dt$$

$$= \frac{1}{2} L \left(\frac{V}{R_2} \right)^2$$

Now, $\Delta Q = \Delta U + \Delta W$

$$\Delta U = 0$$

[$\therefore$ Gas is at constant temperature]

$$\Rightarrow \Delta Q = \Delta W$$

$$= nRT \ln 2 \quad [\therefore P \text{ gets halved}]$$

$$\Rightarrow nRT \ln 2 = \frac{V^2 L}{2R_2^2}$$

$$T = \frac{V^2 L}{nR \cdot R_2^2 \log_e 4}$$

3. **Ans. (A)****Sol.** Consider two coils, having respectively N_1 and N_2 turns, which are so close together that the effective flux in one coil is completely linked with the other. If ϕ is the flux in each turn of (1) when the current is i_1 , the linkages for coil (1) number $N_1 \phi$. The self inductance of this coil is therefore

$$\ell_1 = \frac{N_1 \phi}{i_1}. \text{ The mutual inductance is}$$

$$\frac{N_2 \phi}{i_1} = m = \frac{N_2}{N_1} \ell_1.$$

Similarly, if i_2 is the current in (2) and ϕ'

$$\text{the flux, } \ell_2 = \frac{N_2 \phi'}{i_2} \text{ and } m = \frac{N_1 \phi'}{i_2} = \frac{N_1}{N_2} \ell_2.$$

$$\therefore m^2 = \ell_1 \ell_2 \text{ or } M = \sqrt{L_1 L_2}$$

4. **Ans. (A)****Sol.** The primary rms current

$$I_1 = \frac{P_{av}}{V_1} = \frac{960}{120} = 8$$

$$I_1 = 8A$$

$$\text{Secondary rms current is } I_2 = \frac{P_{av}}{V_2} = \frac{960}{240} = 4$$

5. **Ans. (A)****Sol.** $U_C = 3U_L$

$$U_C + U_L = \frac{1}{2} \frac{Q_0^2}{C}$$

$$4U_L = \frac{1}{2} \frac{Q_0^2}{C}$$

$$\frac{1}{2} LI^2 = U_L = \frac{1}{4} \left(\frac{Q_0^2}{2C} \right)$$

$$I^2 = \frac{1}{4} \frac{Q_0^2}{LC}$$

$$I = \frac{Q_0}{2\sqrt{LC}}$$

6. Ans. (A)

Sol. $P_{av} = \frac{V_0 I_0}{2} \cos \phi_0$

RC : $\tan \phi_{01} = \frac{1}{\omega CR} = \frac{1}{2\pi(500)(4 \times 2) \times 10^{-6}}$

$\tan \phi_{01} = \frac{1}{2\pi \times 10^{-4} \times 40} = \frac{1}{2\pi \times 4 \times 10^{-3}} = \frac{250}{2\pi}$

RL : $\tan \phi_{02} = \frac{\omega L}{R} = \frac{2\pi(500)(1)}{2} = 500\pi$

LC : $\phi_{03} = \frac{\pi}{2}$

$\phi_{01} < \phi_{02} < \phi_{03}$

$(P_{av})_1 > (P_{av})_2 > (P_{av})_3$

7. Ans. (C)

Sol. $V_3 = \sqrt{V_1^2 + V_2^2}$

8. Ans. (D)

Sol. $V_C = V_0 = i_0 X_C$

$V_0 = \frac{i_0}{\omega C} = \frac{V}{\omega CZ}$

$V_0 = \frac{V}{\omega C \sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2}}$

ω is large

$\frac{V_0}{V} \propto \omega^{-2}$

9. Ans. (C)

Sol. We know that voltage across the capacitor in resonance conditions equal to Q times the source voltage = $10 \times 100 = 1000V$.

10. Ans. (A)

Sol. $y = \frac{2 \times 10^{-3}}{(x-2t)^2 + 1}$

$t = 0$

$y = \frac{2 \times 10^{-3}}{x^2 + 1}$

$x = 0, y = 2 \times 10^{-3}$

$x = \pm \infty, y = 0$

Pulse is symmetric about y - axis

11. Ans. (C)

Sol. $\frac{A_R}{A_i} = \frac{\sqrt{\mu_1} - \sqrt{\mu_2}}{\sqrt{\mu_1} + \sqrt{\mu_2}}$

12. Ans. (C)

Sol. $i = i_0 \cos(\omega t)$

[Given at $t = 0, i = i_0$]

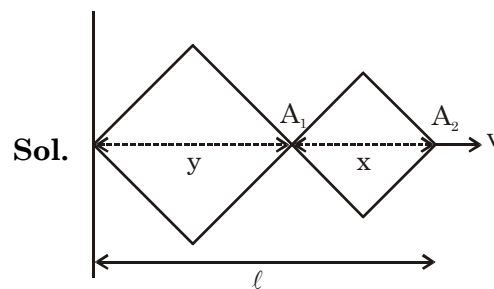
Voltage lags current in case of capacitor

$V \Rightarrow i_0 X_C \cos\left(\omega t - \frac{\pi}{2}\right) [\omega = 2\pi f]$

$\Rightarrow \frac{i_0}{\omega C} \cos\left(\omega t - \frac{\pi}{2}\right)$

$\Rightarrow 50 \sin(\omega t)$

13. Ans. (B)



Sol.

$\ell = y + x$

$\frac{d\ell}{dt} = v = \frac{dy}{dt} + \frac{dx}{dt}$

$\Rightarrow \text{also } \frac{y}{x} = \frac{5}{4} \Rightarrow y = \frac{5}{4}x$

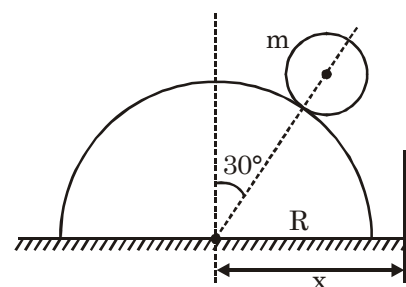
$\frac{dy}{dt} = \frac{5}{4} \frac{dx}{dt} \Rightarrow \frac{dx}{dt} = \frac{4}{5} \frac{dy}{dt}$

$\Rightarrow v = \frac{4}{5} \frac{dy}{dt} + \frac{dy}{dt}$

$\Rightarrow v = \frac{9}{5} \frac{dy}{dt} \Rightarrow \frac{dy}{dt} = \frac{5}{9} v$

14. Ans. (C)

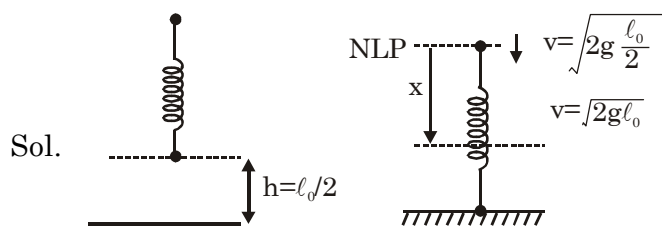
Sol. $0 = \frac{2mx + m[x - (R+r) \sin 30^\circ]}{3m}$



$0 = 3x - (R + r) \sin 30^\circ$

$x = \frac{R + r}{6}$

15. Ans. (D)



$$mgx + \frac{1}{2}k[0^2 - x^2] = 0 - \frac{1}{2}m(g\ell_0)$$

$$mgx - \frac{1}{2} \times \frac{mg}{\ell_0} x^2 = -\frac{mg\ell_0}{2} \quad \left\{ \ell_0 = \frac{mg}{k} \right.$$

$$x - \frac{x^2}{2\ell_0} = -\frac{\ell_0}{2}$$

$$\frac{2\ell_0 x - x^2}{2\ell_0} = -\frac{\ell_0}{2}$$

$$x^2 - 2\ell_0 x - \ell_0^2 = 0$$

$$x = \frac{+2\ell_0 \pm \sqrt{4\ell_0^2 - 4 \times (1) \times (-\ell_0^2)}}{2}$$

$$= \frac{2\ell_0 \pm \sqrt{8\ell_0^2}}{2} \Rightarrow \ell_0 \pm \sqrt{2}\ell_0$$

$$\ell_0 + \sqrt{2}\ell_0 = 2.4\ell_0$$

16. Ans. (A)

Sol. Here the collar velocity can be resolved into two components, one along the slider & other perpendicular to it, let us represent these components by v_r & v_θ respectively.

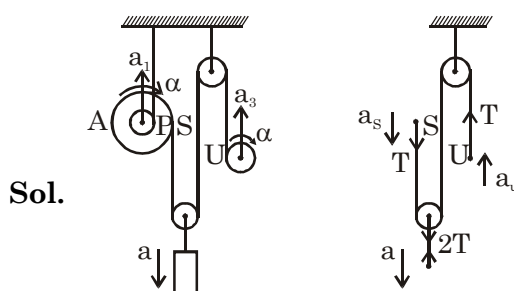
$$v_r = \frac{dr}{dt} = -0.8t \quad \& \quad v_\theta = r \frac{d\theta}{dt} = (3 - 0.4t^2) \times t$$

$$v = \sqrt{v_r^2 + v_\theta^2}$$

$$= \sqrt{(-0.8t)^2 + [(3 - 0.4t^2)t]^2}$$

$$= \sqrt{0.64 + 2.6^2} = \sqrt{7.4} \approx 3$$

17. Ans. (C)



$$a_p = a_1 - \alpha r = a_Q = 0 \Rightarrow a_1 = \alpha r$$

$$\therefore a_U = a_3 + \alpha(2r) = a_3 + 2a_1 \downarrow$$

$$\text{Also, } a_s = \alpha(3r) - a_1 = 2a_1 \downarrow$$

$$\therefore T(a_s) + T(a_U) - 2Ta_2 = 0$$

$$\Rightarrow 2a_2 - a_3 = 4a_1$$

18. Ans. (A)

Sol. $\left(\chi_m \propto \frac{1}{T}\right)$ for paramagnetic material

19. Ans. (D)

$$\text{Sol. } \cos^2 \theta_1 + \cos^2 \theta_2 = \cot^2 \theta_0$$

$$\left(\frac{1}{\sqrt{3}}\right)^2 + \cot^2 \theta_2 = (\sqrt{3})^2$$

$$\cot^2 \theta_0 = 3 - \frac{1}{3}$$

$$\cot \theta_2 = \sqrt{\frac{8}{3}}$$

20. Ans. (D)

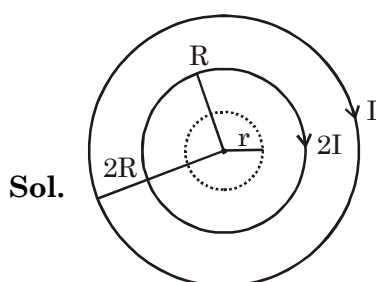
$$\text{Sol. } C\theta = \tau = NiAB$$

$$\Rightarrow \frac{\theta}{i} = \frac{NAB}{C}$$

$$\frac{N_2}{N_1} \times \frac{A_2}{A_1} \times \frac{B_2}{B_1} = 1.4$$

SECTION-II

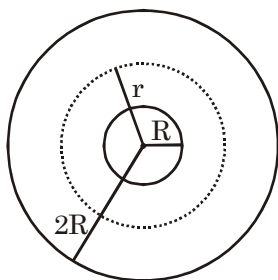
1. Ans. 3.00



$$r < R$$

$$B = \mu_0 n(2I) + \mu_0 nI$$

$$B = 3\mu_0 nI$$



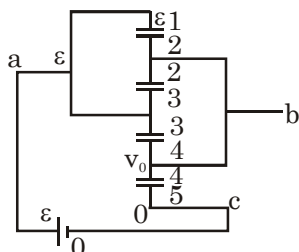
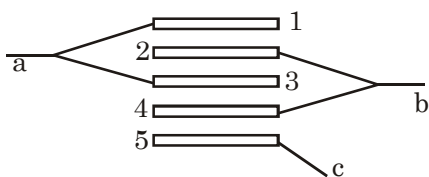
$$R < r < 2R$$

$$B' = \mu_0 nI$$

$$\frac{C}{C''} = 4$$

2. Ans. 0.75

Sol.



$$(v_0 - \varepsilon) + (v_0 - \varepsilon) + (v_0 - \varepsilon) + (v_0 - 0) = 0$$

$$v_0 = \frac{4\varepsilon}{5}$$

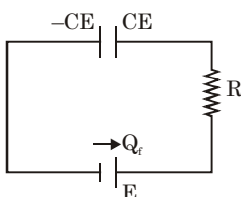
$$-Q = (0 - v_0)C$$

$$-(C_{eq})\varepsilon = -\frac{4\varepsilon}{5}C$$

$$C_{eq} = \frac{4}{5}D = \frac{4\varepsilon_1 B}{5e}$$

3. Ans. 2.00

Sol.



$$\text{Charge flown : } Q_f = 2CE$$

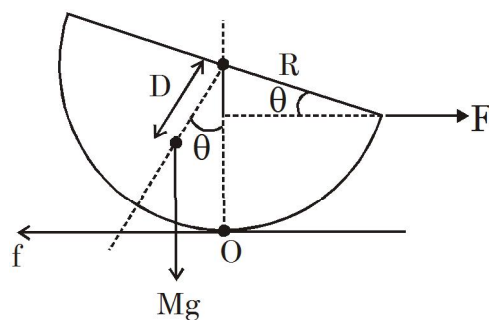
$$W_{batt} = EQ_f = 2CE^2$$

$$\Delta U = U_f - U_i = 0$$

$$W_{batt} = \Delta U + H$$

$$2CE^2 = H$$

4. Ans. 6.00



Sol.

$$D \Rightarrow \frac{4R}{3\pi}$$

$$f \Rightarrow F = \mu mg$$

Balance torque of F and Mg about 'O'.

$$F(R - R\sin\theta) = Mg D \sin\theta$$

$$\mu MgR(1 - \sin\theta) = mg \left(\frac{4R}{3\pi} \right) \sin\theta$$

$$1 - \sin\theta = \sin\theta$$

$$\sin\theta = \frac{1}{2}$$

$$\theta = 30^\circ = \frac{\pi}{6}$$

5. Ans. 9.00

Sol. $\Delta P \Rightarrow P_p - P_i$ will give direction of normal.

$$\Rightarrow -3\hat{i} + 4\hat{j}$$

$$\therefore \text{unit vector } \frac{-3\hat{i} + 4\hat{j}}{5}$$

$$e \Rightarrow \frac{v_{sep}}{v_{app}}$$

$$\Rightarrow \frac{\text{Component of } (\hat{i} + 3\hat{j}) \text{ along normal}}{\text{Component of } (4\hat{i} - \hat{j}) \text{ along normal}}$$

$$e \Rightarrow \left| \frac{-3 + 12}{-12 - 4} \right| \Rightarrow \frac{9}{16}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans.(B)

2. Ans.(D)

3. Ans.(A)

4. Ans.(B)

5. Ans.(B)

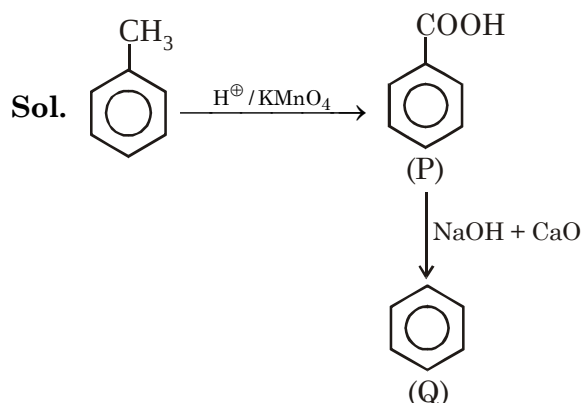
6. Ans.(D)

7. Ans.(C)

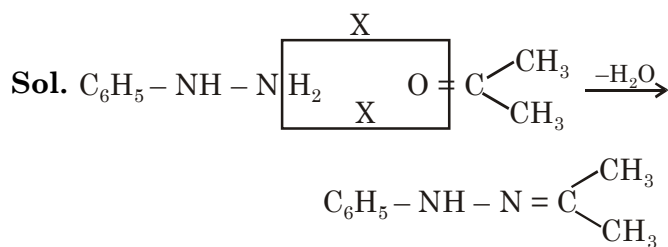
8. Ans.(D)

Sol. $\therefore$ All will form from 1° amine and give Hoffmann mustard oil test.

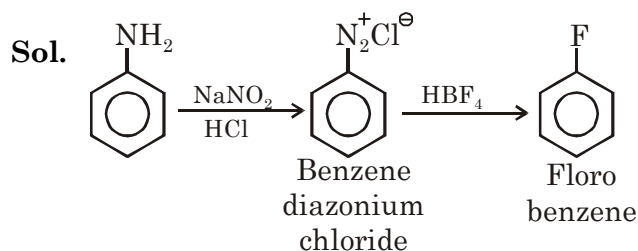
9. Ans. (D)



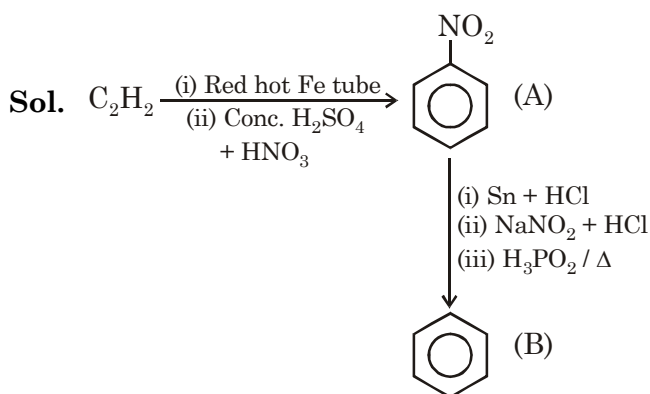
10. Ans. (A)



11. Ans. (D)

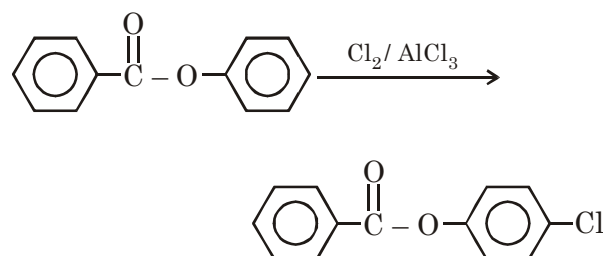


12. Ans. (B)



13. Ans. (C)

Sol.



14. Ans.(C)

15. Ans.(B)

16. Ans. (A)

$$K_{sp} = 108 S^5$$

$$S = \left(\frac{10^{-13}}{108} \right)^{\frac{1}{5}}$$

17. Ans.(D)

18. Ans.(D)

$$\frac{t_2}{t_1} = \left(\frac{a_1}{a_2} \right)^{n-1}$$

19. Ans.(C)

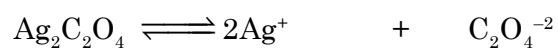
20. Ans.(B)

Sol. $1.5 \times 10^{-3} \times (10-2x) = 0.03 \times 5 \times \frac{40}{1000}$

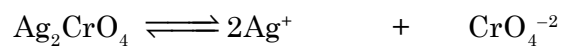
$x = 3$

SECTION-II

1. Ans.(27.00)



$$2x + 2y \quad x$$



$$2y + 2x \quad y$$

$$\frac{x}{y} = \frac{1}{2}, \text{ Also } (2x + 2y)^2 \times x = 10^{-12}$$

$$36 x^3 = 10^{-12}$$

$$x = 0.3 \times 10^{-4}$$

$$y = 0.6 \times 10^{-4}$$

2. Ans.(5.00)

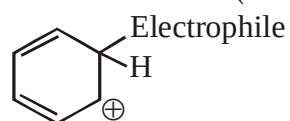
3. Ans. (6.00)

4. Ans. (3.00)

Sol. (iii), (v), (vi)

5. Ans.(3.00)

Reaction (a),(c) & (d) involves wheland's intermediate (s-complex) formation.



PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (B)**

Sol. $(1+x)(1-x)^{2020}$
 $= (1+x)[{}^{2020}C_0 - {}^{2020}C_1x + \dots + {}^{2020}C_{2020}x^{2020}]$
 Required co-efficient
 $= {}^{2020}C_{2020} - {}^{2020}C_1 = 1 - 2020 = -2019$

2. **Ans. (A)**

Sol. $\vec{N} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix}$

$$= -\hat{i} - 7\hat{j} + 5\hat{k}$$

$$|\vec{N}| = \sqrt{1+49+25} = 5\sqrt{3}$$

3. **Ans. (A)**

Sol. $[(2\vec{a} \times 3\vec{b}) \quad (3\vec{b} \times 4\vec{c}) \quad (4\vec{c} \times 2\vec{a})]$
 $= 6 \times 12 \times 8 [\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}]$
 $= 6 \times 12 \times 8 [\vec{a} \quad \vec{b} \quad \vec{c}]^2$
 $= 6 \times 12 \times 8 \times 16 = 2^{10} \times 3^2$

4. **Ans. (C)**

Sol. $|\vec{a} - \vec{b}|^2 = 1 + 1 - 2\vec{a} \cdot \vec{b}$
 $|\vec{b} - \vec{c}|^2 = 1 + 1 - 2\vec{b} \cdot \vec{c}$
 $|\vec{c} - \vec{a}|^2 = 1 + 1 - 2\vec{c} \cdot \vec{a}$
 $|\vec{a} - \vec{b}|^2 + |\vec{b} - \vec{c}|^2 + |\vec{c} - \vec{a}|^2$
 $= 6 - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$
 $= 6 + 3 - |\vec{a}|^2 - |\vec{b}|^2 - |\vec{c}|^2 - 2(\vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{c} + \vec{c} \cdot \vec{a})$
 $9 - |\vec{a} + \vec{b} + \vec{c}|^2 \leq 9$

5. **Ans. (C)**

Sol. Equation of the plane $x + 7y - 5z = -1 - 14 + 5$
 $\Rightarrow x + 7y - 5z + 10 = 0$

$$\text{Required distance} = \frac{10}{\sqrt{1+49+25}}$$

$$= \frac{10}{5\sqrt{3}} = \frac{2}{\sqrt{3}}$$

6. **Ans. (C)**

Sol. Given lines are co-planar

$$\Rightarrow \begin{vmatrix} 2 & 0 & 0 \\ 2 & k & 2 \\ 5 & 2 & k \end{vmatrix} = 0$$

$$\Rightarrow k = 2, (k = -2 \text{ is rejected})$$

Given lines

$$\frac{x-1}{2} = \frac{y+1}{2} = \frac{z}{2} \text{ \& } \frac{x-1}{5} = \frac{y+1}{2} = \frac{z}{2}$$

plane $y + 1 = z$ is common to both the lines

$$\Rightarrow \text{Required plane is } y - z + 1 = 0$$

7. **Ans. (B)**

Sol. Length of the edges are

$$40 - 1, 50 - 2, 60 - 3$$

8. **Ans. (C)**

Sol. Differentiate w.r. to x

$$c \times 2(y + c) \frac{dy}{dx} = 3x^2 \text{ on eliminating } c$$

$$\text{we get } \left(\frac{2x}{3} \left(\frac{dy}{dx} \right) - y \right) \left(\frac{2x}{3} \frac{dy}{dx} \right)^2 = x^3$$

$$\text{order} = 1, \text{degree} = 3$$

9. **Ans. (B)**

Sol. $\frac{dy}{dx} = \left(\frac{y}{x} \right) + \frac{1}{2} \left(\frac{y}{x} \right)^2$

$$\text{put } y = v \cdot x \Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow v + x \frac{dv}{dx} = v + \frac{1}{2} v^2$$

$$\Rightarrow \int \frac{2dv}{v^2} = \int \frac{dx}{x} \Rightarrow 2 \left(\frac{-1}{v} \right) = \ln x + c$$

$$\Rightarrow -2 \frac{x}{y} = \ln x + c$$

$$\text{put } x = 1, y = 2$$

$$c = -1$$

$$\Rightarrow \frac{-2x}{y} = \ln x - 1$$

$$\text{put } x = \frac{1}{4}$$

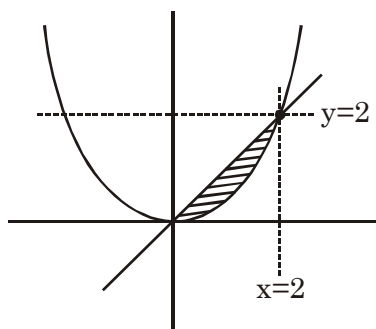
$$-\frac{2}{4xy} = \ln \left(\frac{1}{4} \right) - 1$$

$$\Rightarrow y = \frac{-1}{2 \left(\ln \frac{1}{4} - 1 \right)} = \frac{1}{2 \left(1 - \ln \frac{1}{4} \right)}$$

$$= \frac{1}{2(1 + \ln 4)}$$

10. Ans. (B)

Sol.



$$= \int_0^2 (2x - x^2) dx = x^2 - \frac{x^3}{3} \Big|_0^2$$

$$= 4 - \frac{8}{3} = \frac{4}{3}$$

11. Ans. (B)

Sol. $f(x) = \frac{1}{(x+2019)} \ln(x+2020)$

$$f'(x) = \frac{\frac{x+2019}{x+2020} - \ln(x+2020)}{(x+2019)^2}$$

$$= \frac{x+2019 - (x+2020) \ln(x+2020)}{(x+2020)(x+2019)^2}$$

suppose $h(x) = x+2019 - (x+2020) \ln(x+2020)$

$$\Rightarrow h'(x) = 1 - 1 - \ln(x+2020)$$

$$= -\ln(x+2020)$$

$$h'(x) > 0, \forall x \in (-2020, -2019)$$

$$h'(x) < 0, \forall x \in (-2019, \infty)$$

$$h(-2019) = 0$$

$$\Rightarrow h(x) < 0, \forall x \in (-2020, \infty)$$

$$\Rightarrow f'(x) < 0, \forall x \in (-2020, \infty)$$

12. Ans. (A)

Sol. $f(-1^\oplus) = 4$

$$f(-1) = k + 1$$

$$k + 1 \leq 4$$

$$\Rightarrow k \leq 3$$

13. Ans. (B)

Sol. $f'(x) = 3x^2 - 15x + 18$
 $= 3(x^2 - 5x + 6)$

$$\begin{array}{ccc} f'(x) > 0 & f'(x) < 0 & f'(x) > 0 \\ \hline & 2 & 3 \end{array}$$

$$x^2 - 4x \leq 0 \Rightarrow 0 \leq x \leq 4$$

$$f(x)$$

$$\begin{array}{ccc} \text{Increases} & \text{Decreases} & \text{Increases} \\ \hline 0 & 2 & 3 & 4 \end{array}$$

$$f(2) = 8 - 30 + 36 + 2020$$

$$= 2034$$

$$f(4) = 64 - 120 + 72 + 2020 = 2036$$

14. Ans. (B)

Sol. $\lim_{x \rightarrow 0} \frac{\cos 3x - \cos 2x}{x^2} = \frac{-3 \sin 3x + 2 \sin 2x}{2x}$

$$= \frac{-3}{2} \times 3 + \frac{2}{2} \times 2 = \frac{-9+4}{2}$$

$$= \frac{-5}{2}$$

15. Ans. (A)

Sol. As $f(x)$ must be a constant function

$$\Rightarrow f(x) = \pi$$

$$\sin(\pi) = 0$$

16. Ans. (C)

Sol. $f(x) = \cos x + (x-1) \sin(x-1)$

$$\Rightarrow \text{Differentiable for all } x \in \mathbb{R}$$

17. Ans. (B)

Sol. Suppose $I = \int_0^\pi e^t (\sin^4 t + \cos^4 t) dt$

$$\int_0^{2\pi} e^t (\sin^4 t + \cos^4 t) dt$$

$$= \int_0^\pi e^t (\sin^4 t + \cos^4 t) dt + \int_\pi^{2\pi} e^t (\sin^4 t + \cos^4 t) dt$$

$$= I + \int_{\pi-\pi}^{2\pi-\pi} e^{t+\pi} (\sin^4(t+\pi) + \cos^4(t+\pi)) dt$$

$$= I + e^\pi \int_0^\pi e^t (\sin^4 t + \cos^4 t) dt = I + e^\pi I$$

$$= I(1 + e^\pi)$$

18. Ans. (C)

Sol. $I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x \sin x}{1 + e^x} dx$

$$I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{(-x)(-\sin x)}{1 + e^x} dx$$

$$I + I = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (x \sin x) dx = 2 \int_0^{\frac{\pi}{2}} (x \sin x) dx$$

$$\Rightarrow I = -x \cdot \cos x \Big|_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} \cos x dx$$

$$= 0 + 1$$

19. Ans. (C)

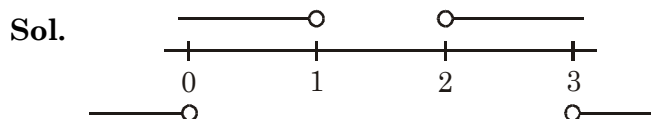
Sol. $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$$B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \Rightarrow B^2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$A = I + B$$

$$A^{2020} = I + 2020B + 0 + 0$$

20. Ans. (D)



$$\Rightarrow a < 0 \text{ or } 3 < a$$

SECTION-II

1. Ans. 1.00

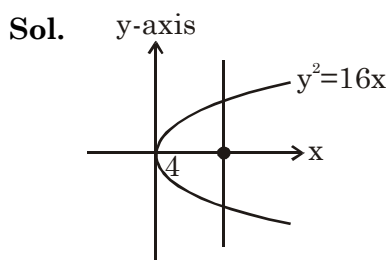
Sol. $2^{2020} = (2^4)^{505} = (1 + 15)^{505}$
 $= 1 + 15 \times 1$

2. Ans. 64.00

Sol. $|\text{adj}(\text{adj } A)| = |\text{adj } A|^4$
 $= |A|^{16} = 2^{16}$

$$\frac{2^{16}}{2^{10}} = 2^6 = 64$$

3. Ans. 0.25



Hence focal chord is latus rectum

Length of latus rectum = 16

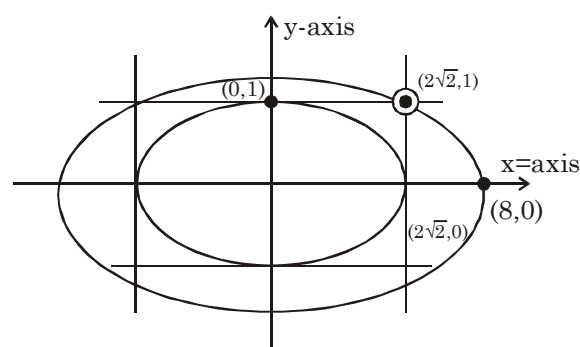
$$\Rightarrow \frac{4}{16} = 0.25$$

4. Ans. 3.00

Sol. Slope of the normal = -1
 foot of normal $(am^2, -2am)$
 $\equiv (1, 2)$
 $(1, 2)$ lies on $x + y = k$
 $\Rightarrow k = 3$

5. Ans. 3.50

Sol.



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

passing through $(2\sqrt{2}, 1), (8, 0)$

$$\Rightarrow \frac{8}{a^2} + \frac{1}{b^2} = 1$$

$$\frac{64}{a^2} + 0 = 1 \Rightarrow a^2 = 64$$

$$\Rightarrow b^2 = \frac{8}{7}$$

$$L = L.R. = \frac{2b^2}{a} = \frac{2 \times 8}{7 \times 8} = \frac{2}{7}$$

$$\Rightarrow \frac{1}{L} = \frac{7}{2} = 3.50$$